

1. K-Means Formula

K-Means aims to minimize the **Within-Cluster Sum of Squares (WCSS)**. The objective function is:

$$J = \sum_{i=1}^N \sum_{k=1}^K z_{i,k} x_i - \mu_k^2$$

Where:

x_i : i -th data point (dimension D).

μ_k : Centroid of cluster k .

$z_{i,k} \in \{0,1\}$: Binary indicator (1 if x_i belongs to cluster k , else 0).

$\therefore$ Euclidean distance:

$$d(a, b) = \sqrt{\sum_{j=1}^D (a_j - b_j)^2}$$

2. Algorithm Steps

K-Means iteratively optimizes J :

1. **Initialize:** Randomly select K initial centroids (e.g., choose points A1, A4, A7).
2. **Assign Points:** Assign each point to the nearest centroid's cluster:

$$\mathcal{C}_k = \{i: k = \operatorname{argmin}_k x_i - \mu_k^2\}$$

3. **Update Centroids:** Recalculate centroids as the mean of assigned points:

$$\mu_k = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} x_i$$

4. **Check Convergence:** Stop if centroids stabilize or max iterations reached; else repeat Steps 2–5.

3. Numerical Examples

Example 1: 1D Data (Simplified)

Data: $\{1, 2, 5, 6, 7\}$, $K = 2$ clusters.

Initial centroids: $\mu_1 = 5$, $\mu_2 = 6$.

Iteration 1:

Assign:

Cluster 1 (near $\mu_1 = 5$): $\{1, 2, 5\} \rightarrow$ New $\mu_1 = (1 + 2 + 5)/3 = 2.67$

Cluster 2 (near $\mu_2 = 6$): $\{6, 7\} \rightarrow$ New $\mu_2 = (6 + 7)/2 = 6.5$

Update: Centroids = $(2.67, 6.5)$.

Iteration 2:

Assign:

Cluster 1 (near 2.67): $\{1, 2\} \rightarrow$ New $\mu_1 = (1 + 2)/2 = 1.5$

Cluster 2 (near 6.5): $\{5, 6, 7\} \rightarrow$ New $\mu_2 = (5 + 6 + 7)/3 = 6$

Update: Centroids = $(1.5, 6)$.

Iteration 3: No changes $\rightarrow$ **Converged**.

WCSS (J):

$$J = (1 - 1.5)^2 + (2 - 1.5)^2 + (5 - 6)^2 + (6 - 6)^2 + (7 - 6)^2 = 0.25 + 0.25 + 1 + 0 + 1 = 2.5$$

Example 2: 2D Data (Detailed)

Data: A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9).

Initial centroids: A1(2,10), A4(5,8), A7(1,2).

Iteration 1:

Assign points (using Euclidean distance):

Point	Dist to A1	Dist to A4	Dist to A7	Cluster
A1	0	3.61	8.06	1
A2	5	4.24	3.16	3
A3	8.49	5.00	7.28	2
A4	3.61	0	7.21	2
A5	7.07	3.16	6.08	2
A6	7.21	4.12	5.10	2
A7	8.06	7.21	0	3
A8	3.61	1.41	7.28	2

- **Clusters:**

Cluster 1: {A1}

Cluster 2: {A3, A4, A5, A6, A8}

Cluster 3: {A2, A7}

Update centroids:

Cluster 1: (2, 10) (unchanged).

$$\text{Cluster 2: } \left(\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5} \right) = (6.0, 6.0)$$

Cluster 3: $\left(\frac{2+1}{2}, \frac{5+2}{2}\right) = (1.5, 3.5)$

Iteration 2+: Reassign points using new centroids until stable.

4. Key Notes

Convergence: May reach local optima (sensitive to initial centroids). Use K-means++ for better initialization.

Choosing K : Apply the **Elbow Method**—plot J (inertia) vs. K and select the "elbow" point.

Limitations: Works only for numerical data; assumes convex, isotropic clusters.

This covers the core mathematics, iterative process, and practical application of K-Means.